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***Abstract*—Reported accuracies for data-driven motor fault diagnosis are frequently near ceiling, yet they rarely measure the ability that industrial deployment actually requires: recognizing the elementary fault mechanisms of a compound fault whose exact combination was never observed during model development. This paper evaluates that ability on a public multimodal dataset of a 2.2-kW three-phase induction motor containing 24 diagnostic states, including nine compound faults that pair a mechanical bearing defect with a second mechanical, electrical, or electromagnetic mechanism, under twelve speed/load conditions. The problem is formulated at the constituent level: nine elementary mechanisms are detected independently from physics-guided, mechanism-specific vibration- and current-domain features, so unseen combinations remain expressible. A crossed protocol separates three levels of compound-context exposure (recombination, one-sided, and two-sided context zero-shot) from operating-condition novelty, with run-level, severity-aware splits and all calibration confined to training data. Although closed-set 24-class accuracy reaches 0.913, exact recovery of unseen constituent sets remains between 0.083 and 0.157 for the locked linear probe. Mechanism-resolved controls show that these failures are strongly heterogeneous. Dynamic eccentricity is almost perfectly discriminable in isolation (AUROC 0.9995; $F_1 = 0.923$), yet its compound recall falls to 0.017 and its compound F_1 to 0.028. Static eccentricity retains substantial discriminative information but shows marked classifier-family sensitivity, whereas winding is already less specific in isolation and remains difficult across classifier families. The primary context-by-condition interaction is positive but becomes unresolved after matching total training-set cardinality. Cross-partner transfer is also heterogeneous and includes one statistically supported score-orientation reversal. These results show that constituent-level evaluation exposes mechanism-specific generalization failures that closed-set accuracy alone conceals.**

***Index Terms*—compound faults, condition monitoring, constituent-level diagnosis, data leakage, electrical machines, evaluation protocols, fault diagnosis, induction motors, multi-label classification, zero-shot learning**

Beyond Closed-Set Accuracy: Constituent-Level Generalization in Compound Motor Fault Diagnosis

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I. INTRODUCTION

INDUCTION motors underpin industrial drivetrains, and their unplanned failures propagate directly into downtime and safety risk. Data-driven condition monitoring has consequently produced a large literature whose reported accuracies now cluster near ceiling on standard laboratory benchmarks. Deployment experience is harder to reconcile with those numbers: a model commissioned at one operating point degrades at another, and faults in service rarely arrive one at a time. A bearing defect that develops alongside a winding short circuit, broken rotor bars, or eccentricity confronts the monitoring system with a *compound* state whose exact combination was, in general, never present in the training history.

Conventional closed-set formulations cannot even express this situation: a previously unseen combination $A+B$ constitutes an unseen class. The compound-fault literature therefore moved toward decoupling formulations that predict elementary mechanisms rather than combinations, and, more recently, toward zero-shot settings in which compound states must be recognized from single-fault training data alone. As Section II reviews, however, that literature has evaluated almost exclusively *homogeneous* compounds—bearing–bearing or bearing–gear pairings whose constituents are all observable in the same vibration channel—and its reported accuracies of roughly 75–87% are not directly comparable with a leakage-safe crossed run-level evaluation, because many representative studies use steady operating conditions and sample- or segment-level evaluation protocols.

This paper asks a deliberately harder question on a public dataset that makes it answerable. The MCC5-THU motor dataset [1] provides 288 recordings of a 2.2-kW three-phase induction motor across 24 diagnostic states under twelve speed/load conditions, with synchronized triaxial vibration, three-phase currents, torque, and a key-phase tachometer. Its nine compound faults include bearing–bearing and bearing-plus-non-bearing combinations involving broken rotor bars, static or dynamic eccentricity, and a stator winding short circuit. Their constituent signatures span vibration- and current-domain evidence, providing a heterogeneous multimodal setting for constituent-level generalization. To our knowledge, neither this cross-modal compound regime nor this dataset has previously received a constituent-level evaluation.

The study is designed so that every claim is bounded by an explicit control. The formulation is constituent-level: nine elementary mechanisms are detected independently by regularized linear probes operating on curated, mechanism-specific physical-signature families derived from angle-resampled order spectra, envelope-order descriptors, and phase-current sequence measures. A crossed protocol varies compound-context exposure over three regimes—recombination, one-sided, and two-sided compound-context zero-shot—and crosses each with seen and unseen operating conditions, under run-level, severity-aware splits with all preprocessing, calibration, and threshold selection confined to training data. The principal constituent-level metrics are accompanied by input-independent constant-predictor references, so that prevalence-driven scores cannot masquerade as diagnostic skill; oracle detectability analyses are paired with recording-sensitivity and same-date controls, so that separability is not silently attributed to fault mechanisms; and paired run-clustered bootstrap inference, supplemented by composition-level cluster sensitivity analysis for the context-by-condition interaction, distinguishes supported effects from sampling-unit sensitivity.

This paper makes four contributions. First, it introduces a constituent-level formulation and a crossed composition–context–condition protocol for compound motor faults whose constituents span mechanical and electrical modalities, separating novel composition, compound-context deprivation, and operating-condition novelty within a single design (Sections III–IV). Second, it develops a prior-aware, mechanism-resolved diagnostic evaluation that combines input-independent prevalence references, constituent-level ranking and thresholded performance, isolated-fault learnability controls, and classifier-family sensitivity. This decomposition helps distinguish empirical patterns consistent with weak constituent specificity, classifier-capacity sensitivity, or degradation that emerges when an otherwise learnable constituent is evaluated in compound faults. Third, it presents a controlled analysis of constituent transfer, spanning within-pair oracle detectability, cross-partner transfer, physical-signature preservation, and a statistically supported score-orientation reversal, each bounded by recording-sensitivity and same-date controls that reduce the risk of reading recording-related variation as fault physics. Finally, sensitivity and robustness analyses over the decision-threshold objective, single-fault severity merging, and four classifier families show that absolute decomposition quality is sensitive to operating point and classifier family, while the unseen-condition recombination

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advantage over two-sided withholding remains directionally positive across all four classifier families.

The principal findings are that closed-set discriminability (accuracy 0.913) coexists with exact constituent-set recovery of only 0.083–0.157 on unseen compounds for the locked probe; and that the resulting failures are strongly mechanism specific rather than uniform. Bearing-inner, bearing-outer, and broken-bar constituents remain strongly discriminable, whereas dynamic eccentricity undergoes a particularly severe transition from near-perfect isolated-fault discrimination (AU-ROC 0.9995; $F_1 = 0.923$) to compound recall of 0.017 and F_1 of 0.028. Static eccentricity retains substantial discriminative information but is strongly classifier-family sensitive, whereas winding is already less constituent-specific in isolation and remains poor across nonlinear alternatives. Cross-partner evidence is often transferable but can additionally undergo severe context-dependent score reorientation. The primary crossed analysis suggested a larger penalty from complete compound-context deprivation when the operating condition was also unseen, but this interaction became unresolved after matching total training-set cardinality and is therefore treated as a sensitivity-dependent directional observation rather than a training-support-independent effect.

The remainder of the paper is organized as follows. Section II reviews related work and positions the study. Section III details the constituent representation, feature construction, detection, protocol, metrics, and statistical analysis. Section IV reports results and Section V concludes the paper.

II. Related Work

Four research threads intersect in this study: compound-fault decoupling, zero-shot recognition of unseen fault combinations, the reliability of evaluation protocols in data-driven condition monitoring, and generalization across operating conditions. We review each and then position the present work (Table I).

Early data-driven treatments of compound faults assigned each observed combination its own class, which cannot express combinations absent from training. Decoupling formulations address this by predicting the elementary mechanisms instead. Huang *et al.* proposed a deep decoupling convolutional network whose multi-label classifier recovers the constituents of gearbox compound faults from raw vibration signals [2], and a one-dimensional CNN variant with a multi-label output layer [3]. Subsequent work refined the decoupling head: capsule networks with dynamic routing and threshold-based label activation [4], attentional residual networks combined with active learning for measured (rather than injected) bearing compound faults [5], and mixture-of-experts architectures with task-adapted attention for decoupling compounds unseen as combinations [6]. Two properties are common among the representative studies considered here: the constituents are *mechanical* (bearing races, balls, gear teeth), so every constituent is observable in the same vibration channel; and several evaluations rely on steady-speed laboratory recordings and sample- or segment-level protocols that do not explicitly test independence across physical recordings.

A stricter setting withholds all compound data and requires composing single-fault knowledge at test time. Xu *et al.* introduced zero-shot compound-fault diagnosis with manually defined semantic attributes learned from single-fault data [7], later replacing hand-crafted attributes with autoencoder-learned fault semantics [8]. Generative formulations synthesize pseudo-samples of unseen compounds from single-fault semantics [9], [10]; graph embeddings propagate relational structure among base faults [11]; and physics-informed semantic spaces anchor the attribute vectors in characteristic-frequency knowledge [12]. Reported accuracies on unseen compounds cluster between roughly 75% and 87% [7], [8], [12], [13].

These results, however, concern compounds whose constituents are homogeneous: bearing gear combinations in which both mechanisms modulate the same vibration carrier. The MCC5-THU motor dataset instead contains compound compositions that combine bearing defects with broken-bar, eccentricity, and winding mechanisms. Their curated signatures span current-domain carrier and imbalance descriptors as well as, for eccentricity, shaft-order vibration descriptors. The resulting setting therefore tests heterogeneous multimodal constituent transfer rather than a purely vibration-domain compound task. To our knowledge, this cross-modal compound regime has not been evaluated in the zero-shot compound-fault literature, and no published constituent-level results exist for this dataset.

A parallel literature shows that evaluation-protocol choices, more than model choices, drive many reported near-ceiling accuracies. Wheat *et al.* quantified the impact of intra-bearing data leakage in vibration diagnosis and advocated part-to-part splits [14]; recent work formalizes the distinction between segmentation-level and specimen-level leakage and proposes leakage-safe protocols [15], [16]; and open benchmark studies made similar observations for deep architectures across public datasets [17]. This thread has concentrated on single-fault classification. Its intersection with compound-fault evaluation is, to our knowledge, unexplored — yet it is precisely where leakage is most subtle: in the dataset used here, severity variants of one physical combination can leak across a composition boundary, while compound and single-fault recordings may also differ in recording-related conditions, which can confound naive "compound versus single" contrasts. Our protocol design (run-level, severity-aware splits; recording-sensitivity and same-date controls; and constant-predictor references for the principal constituent-level metrics) extends leakage-aware evaluation to the compositional setting.

Diagnosis under variable speed and load has produced a large domain-adaptation and domain-generalization literature, surveyed in [18]. On the MCC5-THU family specifically, the dataset authors proposed evidential ensemble learning for real-time multimode diagnosis [19] and recently questioned how operating conditions should enter learning-based diagnosis at all [20]; other groups applied condition-robust architectures to the companion gearbox release [21], [22]. The dominant protocol in this literature holds one condition out of many [18], [23]. The present study instead evaluates operating-condition novelty jointly with compound-context exposure through the crossed composition–context–condition design.

TABLE I: Positioning of this study against representative prior work. ✓ = explicitly included; – = absent, not applicable, or not reported. "Cross-modal" means the compound's constituents are observable in different sensing modalities (mechanical + electrical). "Prior-aware references" means metrics are reported against label-prevalence (trivial constant-predictor) references.

Study	Task focus	Signals	Compound faults	Cross-modal	Unseen combin.	Condition shift	Leakage-safe run-level splits	Prior-aware references
Huang *et al.* [2]	multi-label decoupling	vib.	✓	–	–	–	–	–
LTSS-BoW-CapsNet [4]	capsule decoupling	vib.	✓	–	–	–	–	–
HeMTAN [6]	unseen-compound decoupling	vib.	✓	–	✓	–	–	–
Xu *et al.* [7]	zero-shot compounds	vib.	✓	–	✓	–	–	–
Xu *et al.* [8]	learned fault semantics	vib.	✓	–	✓	–	–	–
Song *et al.* [12]	GZSL, physics semantics	vib.	✓	–	✓	–	–	–
Wheat *et al.* [14]	leakage in evaluation	vib.	–	–	–	–	✓	–
Han *et al.* [18]	multi-condition survey	multi	–	–	–	✓	–	–
Liu *et al.* [19]	multimode diagnosis	vib.+cur.	–	–	–	✓	–	–
DRMNN [24]	multimodal fusion	vib.+cur.	–	–	–	–	–	–
This work	constituent-level generalization	vib.+cur. (+key-phase)	✓	✓	✓	✓	✓	✓

That mechanical faults imprint on stator currents and electrical faults on vibration has long motivated multimodal motor diagnosis [25]. Deep multimodal fusion typically concatenates modalities into one representation or fuses learned features, e.g. multilevel information fusion [26], dynamic-routing multimodal networks [24], adaptive CNN fusion of denoised vibration and current [27], and comparative studies of the two modalities under deep and classical learners [28]. These works report that multimodal fusion can improve conventional single-fault diagnosis. In the present study, constituent detectors instead use predefined mechanism-specific feature families, restricting each detector to physically relevant vibration- or current-domain descriptors and thereby reducing direct competition between unrelated modality features.

Public rotating-machinery datasets underpin this literature: CWRU [29] and Paderborn [30] for bearings, and more recent multi-fault motor and drivetrain releases such as a PMSM stator-fault dataset [31] and an industrial-scale motor–pump dataset [32]. The MCC5-THU family — a gearbox release [33] and the motor release used here [1] — is distinctive in combining single and compound faults, twelve speed/load profiles with transitional segments, and synchronized vibration, three-phase current, torque, and key-phase channels. The motor release is distributed without constituent-level reference baselines; the present study provides, to our knowledge, its first systematic constituent-level evaluation.

Table I contrasts this study with representative prior work along the axes developed above. The distinguishing combination is: compound faults whose constituents live in different modalities; composition, compound-context, and operating-condition generalization varied jointly in one crossed protocol; run-level, severity-aware, leakage-safe splits; the principal constituent-level metrics interpreted against constituent-specific constant-predictor references so that prior-driven scores cannot masquerade as diagnostic skill; and run-clustered bootstrap inference with multiplicity control.

III. Materials and Methods

The study evaluated constituent-level recovery of compound motor faults under novel composition, compound-context, and operating-condition settings. Rather than treating each compound state as a separate multiclass label, faults were represented by their elementary mechanisms. The analysis combined physics-guided multimodal features, independent constituent detectors, a crossed generalization protocol, transfer and signature-preservation analyses, and statistical evaluation. All preprocessing, normalization, classifier fitting, and threshold calibration were confined to the corresponding training data to prevent information leakage.

A. Dataset and Constituent-Level Problem Formulation

The study used the public MCC5-THU motor dataset [1], comprising 288 physical recordings from a 2.2-kW three-phase induction motor across 24 diagnostic states and 12 speed/load operating conditions. The release contains healthy, single-fault, and nine compound-fault compositions, yielding 108 physical compound runs. Synchronized measurements include triaxial vibration, three-phase current, torque, and keyphase signals; the present constituent analysis used vibration, current, and keyphase measurements. Each experimental run i was represented by a binary constituent vector

$$\mathbf{y}_i = [y_{i,1}, \ldots, y_{i,K}]^{\mathrm{T}}, \qquad \mathbf{y}_i \in \{0,1\}^K, \tag{1}$$

with $K = 9$ elementary mechanisms: bearing ball, bearing inner race, bearing outer race, shaft bending, broken bar, dynamic eccentricity, static eccentricity, voltage unbalance, and winding fault.

The label cardinality distinguished healthy, single-fault, and compound runs,

$$\|\mathbf{y}_i\|_0 = \begin{cases} 0, & \text{healthy}, \\ 1, & \text{single fault}, \\ \geq 2, & \text{compound fault}. \end{cases} \tag{2}$$

The constituent composition of run i was defined as

$$\mathcal{C}_i = \{f_k : y_{i,k} = 1\}. \tag{3}$$

Unlike closed-set multiclass diagnosis, where an unseen combination $A + B$ forms an unseen class, the constituent formulation seeks to recover A and B independently. The resulting generalization problem is therefore whether constituent knowledge learned from single faults and other compound contexts transfers to a novel composition.

B. Physics-Guided Multimodal Feature Representation

Features were derived from three vibration and three phase-current channels sampled at 12,800 Hz. Signals were segmented into 2-s analysis windows using a 1-s stride, corresponding to 50% overlap. Each window was angle resampled from the measured keyphase pulses at 256 samples per shaft revolution before order-domain analysis. For physics-feature extraction, at most 12 windows were retained per recording; when more were available, approximately evenly spaced windows spanning the recording were selected. Window-level descriptors were aggregated to one run-level representation by their median. Although interquartile-range descriptors were also generated, the curated constituent detectors used only the median features and excluded the corresponding `__iqr` terms.

The run-level multimodal representation was

$$\mathbf{x}_i = \left[\left(\mathbf{x}_i^{(v)} \right)^{\mathrm{T}}, \left(\mathbf{x}_i^{(c)} \right)^{\mathrm{T}} \right]^{\mathrm{T}}, \tag{4}$$

where $\mathbf{x}_i^{(v)}$ and $\mathbf{x}_i^{(c)}$ denote vibration- and current-derived descriptors.

For an angle-domain signal $x[n]$, a Hann-window periodogram $P_x(o)$ was computed. Energy around a target shaft order o_0 was accumulated within ± 0.12 order and normalized by the total energy over 0.25–60 orders,

$$R_x(o_0) = \frac{\sum_{|o-o_0|\le 0.12} P_x(o)}{\sum_{0.25\le o\le 60} P_x(o) + \epsilon}. \tag{5}$$

Both local energy and $\log_{10}[\max(R_x(o_0), \epsilon)]$ were retained.

The fixed ± 0.12 half-width was verified against the released signals rather than adopted by convention. Because a fixed absolute window spans a smaller relative width at higher harmonics while slip-induced deviation of a bearing order grows with harmonic number, the observed spectral peak departs from its kinematic order by a median of 0.024 and 0.036 order at the BPFO and BPFI fundamentals but 0.049 and 0.080 order at their second harmonics; the ± 0.12 window consequently contains the observed peak in 96–100% of single-fault runs at the fundamental and in 74% at the BPFI second harmonic. Widening the window does not repair this and degrades discrimination, because $2o_{\mathrm{BPFO}}$ and $2o_{\mathrm{BPFI}}$ lie close to integer shaft orders that carry rotational energy in healthy and faulty machines alike: separation of faulty from healthy runs at $2o_{\mathrm{BPFO}}$ falls from AUROC 0.877 at ± 0.12 to 0.573 at ± 0.25 and to chance at ± 0.50. The ± 0.12 choice is therefore retained as the better available compromise on this rig, with second-harmonic descriptors—BPFI's in particular—understood to carry less reliable evidence than their fundamentals.

For the SKF 6205 bearing, the characteristic orders were

$$o_{\mathrm{BPFO}} = 3.585, \qquad o_{\mathrm{BPFI}} = 5.415, \qquad o_{\mathrm{BSF}} = 2.357, \tag{6}$$

together with their second harmonics. Bearing descriptors were extracted from the Hilbert-envelope vibration order spectrum. Importantly, the Hilbert envelope was computed *after* angle resampling; therefore, these descriptors characterize modulation retained in the 256-samples-per-revolution angle-domain representation rather than a separately band-pass-filtered broadband demodulation signal. The angle-domain Nyquist limit is 128 shaft orders, while the normalized spectral ratios used here were restricted to 0.25–60 orders.

Current descriptors were obtained from the individual phase-current spectra and the Clarke-space-vector magnitude. The dominant electrical carrier o_e was identified from the maximum summed three-phase spectral energy over 0.5–30 orders, and descriptors were evaluated at

$$o_e, \qquad o_e \pm 1, \qquad o_e \pm 2, \qquad 2o_e. \tag{7}$$

The $o_e \pm 1$ and $o_e \pm 2$ terms were treated as carrier-relative broken-bar-sensitive proxies rather than as a fully slip-parameterized broken-bar model. Because these sidebands may also vary with load, operating condition, and other electromagnetic modulation phenomena, they are not interpreted as mechanism-exclusive slip-frequency measurements.

The distinction is quantitative on this rig. The detected carrier lies at 50.06 Hz against a 49.69 Hz shaft at the 3000 rpm setting, identifying a two-pole machine, so that $s = 1 - 1/o_e$ yields slip of 3.0%, 1.5%, and 1.0% at the 1000, 2000, and 3000 rpm settings. The classical broken-bar sidebands at $(1 \pm 2s)f_e$ therefore sit $2s\, o_e = 0.062$, 0.031, and 0.020 order from the carrier—about two spectral bins at the analysis window length used here, and within the ± 0.12 band centered on o_e itself in 92.7% of analyzed windows. That signature is thus absorbed into the carrier descriptor rather than resolved separately, whereas $o_e \pm 1$ and $o_e \pm 2$ lie one and two shaft orders out, coinciding with the $f_e \pm f_r$ spacing associated with eccentricity. Resolving genuine broken-bar sidebands on this rig would require substantially longer analysis windows or explicit slip-tracked demodulation.

Current imbalance was characterized by the coefficient of variation of the three phase RMS values and a phase-order-invariant sequence ratio. For the two sequence candidates S_1 and S_2 at o_e,

$$r_{\mathrm{seq}} = \frac{\min(|S_1|, |S_2|)}{\max(|S_1|, |S_2|) + \epsilon}, \tag{8}$$

which avoids assuming a predefined phase order and therefore represents sequence imbalance rather than a prespecified negative-sequence magnitude.

Each constituent detector received a fixed mechanism-specific subset of this feature bank. Bearing-inner, bearing-outer, and bearing-ball detectors used BPFI/2BPFI, BPFO/2BPFO, and BSF/2BSF envelope-order features, respectively; shaft bending used shaft-1× and shaft-2× vibration features; broken bar used carrier-sideband proxies; dynamic and static eccentricity used shaft-order and carrier-sideband features; voltage unbalance used sequence-ratio and phase-RMS imbalance; and winding used sequence-ratio, phase-

RMS imbalance, and electrical second-harmonic features. Formally,

$$\mathbf{x}_i^{(k)} = \mathbf{M}_k \mathbf{x}_i, \tag{9}$$

where $\mathbf{M}_k$ is the predefined selector for constituent k. The resulting subsets contained 16 descriptors for each bearing mechanism and shaft bending, 40 for broken bar, 56 for each eccentricity mechanism, 2 for voltage unbalance, and 12 for winding. These dimensions arose from the predefined channel–order combinations rather than post-hoc optimization, and no outer-test performance was used to alter the feature definitions or selector sizes.

C. Constituent Detection and Sensitivity Analyses

A separate binary detector was trained for each constituent mechanism, allowing multiple faults to be activated simultaneously without a softmax constraint. The locked primary model used L_2-regularized logistic regression ($C = 1$) with class balancing and the predefined mechanism-specific feature subset $\mathbf{x}_i^{(k)}$ for constituent k.

All preprocessing was fitted within the outer-training fold: missing values were imputed using training-set medians and features were standardized using training-set statistics. Constituent probabilities were converted to binary decisions as

$$\hat{y}_{i,k} = \mathbb{I}\left(p_{i,k} \geq \tau_k\right), \tag{10}$$

where τ_k is a constituent-specific threshold.

Thresholds were selected exclusively from outer-training data using `GroupKFold` grouped by `condition_key`, with four inner folds whenever the available operating-condition groups permitted and fewer folds only when necessary. Each operating condition was confined to a single inner fold. Out-of-fold probabilities from the valid inner splits were pooled for each constituent, and candidate thresholds from 0.05 to 0.95 in increments of 0.01 were ranked primarily by constituent-level F_1, with balanced accuracy and proximity to 0.5 used as deterministic tie breakers. After threshold selection, the constituent detector was refitted on the complete outer-training partition, and the selected threshold was fixed for outer-test evaluation. No outer-test sample contributed to imputation, standardization, classifier fitting, or threshold selection.

As post-hoc operating-point sensitivity analyses, threshold calibration was repeated using $F_{0.5}$ and balanced accuracy as alternative objectives while keeping the outer splits, preprocessing, fitted detectors, grouped inner validation, and threshold grid unchanged. These alternatives were used only to characterize the precision–recall trade-off.

Logistic regression was retained as the locked primary probe to emphasize constituent transferability rather than architecture optimization. Alternative classifier families and single-fault severity definitions were evaluated separately as sensitivity analyses.

Because the primary formulation merges high- and low-severity single-fault examples, a post-hoc analysis tested whether low-severity support affected the crossed generalization results. Low-severity single-fault runs for bearing inner, bearing outer, static eccentricity, and winding were removed (48 runs), reducing the analysis set to 240 runs while leaving all 108 compound evaluation runs unchanged. The constituent representation, curated features, logistic-regression model, crossed outer splits, and grouped inner-CV threshold calibration were otherwise unchanged. Differences from the locked H/L-merged analysis were assessed with 5000 paired physical-run cluster-bootstrap resamples.

The crossed protocol was additionally repeated with a random forest, an RBF-SVM, and a three-hidden-layer MLP as post-hoc robustness references. The random forest used 100 trees, square-root feature subsampling, a minimum leaf size of two, and balanced-subsample class weighting. The RBF-SVM used $C = 1$, `gamma=scale`, and balanced class weighting. The MLP used 64–32–16 hidden units with ReLU activation, Adam optimization, $L_2 = 10^{-4}$, an initial learning rate of 10^{-3}, and a maximum of 400 iterations; class imbalance was handled using balanced sample weights computed from the outer-training labels.

Across classifier families, constituent labels, curated feature sets, outer splits, training-only preprocessing, grouped inner cross-validation, threshold selection, and evaluation metrics were held fixed. No alternative model or hyperparameter configuration was selected using outer-test performance.

To determine whether poor compound recovery reflected constituent prevalence, limited classifier capacity, weak isolated-fault detectability, or degradation specific to compound contexts, two post-hoc diagnostic analyses were performed. First, constituent-level performance was summarized with equal weight across the six crossed evaluation scenarios. For the locked logistic regression, the thresholded constituent metrics were complemented by AUROC computed from the continuous outer-test scores and by an input-independent always-positive F_1 reference for each constituent. Corresponding F_1 values from RF, RBF-SVM, and MLP were used only as descriptive classifier-capacity sensitivities and not for post-hoc model selection.

Second, isolated-fault learnability was evaluated using leave-one-operating-condition-out testing with the same curated feature sets, logistic-regression specification, training-only preprocessing, and grouped inner threshold calibration as the locked pipeline. For each constituent, isolated single-fault runs were contrasted both with healthy runs and with all other non-compound states. All runs from the held-out operating condition were removed before preprocessing, model fitting, and threshold calibration. Compound runs were excluded entirely from this control. The single-fault-versus-other-non-compound contrast was used for the primary isolated-learnability comparison because it tests constituent specificity rather than separation from health alone.

D. Closed-Set Control Experiment

An auxiliary 24-class closed-set experiment was used to quantify conventional in-distribution separability before compositional constraints were imposed. The reference model was a raw-signal multimodal 1-D CNN with separate three-channel vibration and current encoders. Each encoder contained four convolutional blocks (32, 64, 128, and 192 channels), adaptive average pooling, and a 128-dimensional embedding. The two embeddings were concatenated and passed through a 256-unit fusion layer and a 24-class softmax output.

The network used 2-s windows sampled at 12,800 Hz and five-fold stratified cross-validation at the physical-run level, so all windows from a run remained within the same outer fold. Within each outer-training partition, 20% of the training runs were reserved for stratified validation. Channel normalization parameters were estimated from training data only and applied unchanged to validation and test windows.

Training used cross-entropy loss and AdamW ($\mathrm{lr} = 10^{-3}$, weight decay 10^{-4}, dropout 0.25), with a maximum of 25 epochs and early-stopping patience of five. The retained checkpoint maximized validation run-level macro-F_1, with validation loss as the tie breaker. At inference, window-level softmax probabilities were averaged within each run and the highest mean probability defined the run-level class.

Final performance was computed from pooled out-of-fold predictions so that each of the 288 runs was tested exactly once. This experiment served only as an auxiliary closed-set reference and did not influence model or feature selection in the constituent-generalization analyses.

E. Crossed Composition–Context–Condition Generalization Protocol

A crossed outer protocol separated novel composition, compound-context exposure, and operating-condition shift. Compound structure was represented as an undirected graph $G = (V, E)$, where $V = \{f_1, \ldots, f_K\}$ contains the constituent mechanisms and an edge $(A, B) \in E$ denotes an observed compound. For a target composition $\mathcal{C}^* = \{A, B\}$, the three context regimes correspond to increasingly restrictive graph withholding (Fig. 1).

Let $d_{\mathrm{cmp}}(f)$ denote the number of distinct compound contexts containing constituent f in the outer-training set. The regimes were

$$\begin{aligned} \text{Recombination:} &\quad d_{\mathrm{cmp}}(A) > 0,\ d_{\mathrm{cmp}}(B) > 0, \\ \text{One-sided:} &\quad d_{\mathrm{cmp}}(A) = 0,\ d_{\mathrm{cmp}}(B) > 0, \\ \text{Two-sided:} &\quad d_{\mathrm{cmp}}(A) = 0,\ d_{\mathrm{cmp}}(B) = 0. \end{aligned} \tag{11}$$

Recombination withheld only the target pair $A + B$ while retaining both constituents in other compounds. One-sided context zero-shot removed all compound exposure for one target constituent while retaining its eligible single-fault examples; both directions were evaluated. Two-sided context zero-shot removed compound exposure for both target constituents while retaining their eligible single-fault examples.

Each context regime was crossed with operating-condition exposure. Let $\mathcal{O}_{\mathrm{train}}$ denote the set of operating conditions represented in the outer-training data,

$$c_{\mathrm{test}} \in \mathcal{O}_{\mathrm{train}} \quad \text{(seen)}, \qquad c_{\mathrm{test}} \notin \mathcal{O}_{\mathrm{train}} \quad \text{(unseen)}. \tag{12}$$

For unseen-condition evaluation, all runs acquired under c_{test} were removed from the outer-training set before preprocessing, classifier fitting, and threshold calibration. Eligible runs from the target condition remained available in the seen-condition setting. The resulting design comprised three context regimes crossed with two condition regimes, yielding six principal scenarios.

All splits were defined at the physical-run level before model fitting. Composition identity was derived from the constituent vector rather than severity-specific class names, preventing severity variants of the same composition from crossing the withholding boundary. The directional one-sided evaluation produced 216 prediction rows from 108 physical compound runs; recombination and two-sided evaluation each produced 108 rows. The two one-sided predictions from the same run were kept within the same run-level bootstrap cluster. The recombination-versus-two-sided context-by-condition interaction was additionally evaluated using composition-level clustering (Section III-H).

For reproducibility, Algorithm 1 summarizes the outer-split construction. Context and condition withholding always preceded preprocessing and model fitting.

Algorithm 1. Outer-split construction for the crossed protocol.

for each target compound composition $\mathcal{C}^* = \{A, B\}$ **do**
  for each compound-context regime **do**
    if recombination **then**
      remove the target compound $A + B$ from the training pool
    else if one-sided context zero-shot for A **then**
      remove all compound runs containing A
      retain eligible single-fault runs containing A
    else if one-sided context zero-shot for B **then**
      remove all compound runs containing B
      retain eligible single-fault runs containing B
    else if two-sided context zero-shot **then**
      remove all compound runs containing either A or B
      retain eligible single-fault runs containing A or B
    end if
    for each operating-condition regime **do**
      if unseen condition **then**
        remove all outer-training runs acquired under c_{test}
      end if
      fit preprocessing using outer-training data
      fit constituent detectors
      select thresholds by grouped inner cross-validation
      evaluate the held-out target compound run(s)
    end for
  end for
end for

The withholding regimes also produced different amounts of training support. Recombination used 276 and 253 training runs per seen- and unseen-condition fold, respectively, whereas

TABLE II: Outer-split cardinalities under the crossed evaluation protocol. Training-set sizes are reported as minimum/median/-maximum numbers of physical runs across the corresponding outer folds. The one-sided regime is evaluated in both directional forms, producing two prediction rows per physical compound run.

Condition	Context regime	Outer folds	Train runs (min/med/max)	Test runs/fold	Prediction rows
Seen	One-sided	18	228 / 228 / 264	12	216
Seen	Recombination	9	276 / 276 / 276	12	108
Seen	Two-sided	9	180 / 216 / 216	12	108
Unseen	One-sided	216	209 / 209 / 242	1	216
Unseen	Recombination	108	253 / 253 / 253	1	108
Unseen	Two-sided	108	165 / 198 / 198	1	108

the corresponding median two-sided sizes were 216 and 198. Median positive support for the target constituents decreased from 71 to 23.5 under seen conditions and from 65 to 22 under unseen conditions. Thus, context withholding also induces a training-support difference, which is examined separately in a size-matched sensitivity analysis.

To assess whether the recombination advantage could be explained by total training-set cardinality alone, recombination training sets were repeatedly downsampled within operating-condition groups to match the corresponding two-sided training-set sizes. The original test sets, constituent representation, curated features, preprocessing, logistic-regression model, and grouped inner-CV threshold calibration were retained unchanged.

Downsampling was constrained to preserve at least one non-target compound context for each target constituent, so that the defining recombination exposure was maintained. Thirty repeated size-matched subsamples were evaluated. Uncertainty due to both composition sampling and training-set subsampling was summarized using 5000 composition-level bootstrap resamples over the nine target compositions, with a size-matched repeat sampled within each bootstrap replicate. This analysis controls total training-set cardinality and operating-condition distribution, but does not equate target-positive support, which is part of the compound-context exposure difference by design.

F. Evaluation Metrics

Performance was evaluated at the constituent-set level. Constituent recall was

$$R_{\text{const}} = \frac{\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k}\hat{y}_{i,k}}{\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k}}, \tag{13}$$

and exact-match recovery was

$$\text{EM} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(\hat{\mathbf{y}}_i = \mathbf{y}_i\right). \tag{14}$$

False additions per prediction row were quantified as

$$\text{FA/run} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}(1-y_{i,k})\hat{y}_{i,k}. \tag{15}$$

Recovery of at least one and all true constituents was defined as

$$R_{\text{any}} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(\sum_{k=1}^{K} y_{i,k}\hat{y}_{i,k} > 0\right), \tag{16}$$

$$R_{\text{all}} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(\sum_{k=1}^{K} y_{i,k}\hat{y}_{i,k} = \sum_{k=1}^{K} y_{i,k}\right). \tag{17}$$

Micro-F_1 was computed over all constituent decisions,

$$F_{1,\text{micro}} = \frac{2TP}{2TP + FP + FN}. \tag{18}$$

Macro-F_1 was additionally reported over only the constituents represented positively in the corresponding evaluation subset,

$$F_{1,\text{active}} = \frac{1}{|\mathcal{K}_+|}\sum_{k\in\mathcal{K}_+} F_{1,k}, \tag{19}$$

where $\mathcal{K}_+$ contains constituents with at least one positive test instance.

G. Within-Pair Detectability, Cross-Partner Transfer, and Signature Preservation

To distinguish failed constituent recovery from absence of measurable evidence, constituent A in a target compound $A+B$ was first evaluated by the within-pair contrast

$$A + B \quad \text{vs.} \quad B, \tag{20}$$

with discriminability quantified as $\text{AUC}^{\text{within}}_{A|B}$.

Cross-partner transfer then tested whether evidence for A learned from eligible compound contexts other than $A + B$ generalized to the same target contrast. The target pair was excluded from training, and the resulting fixed-orientation score was summarized by $\text{AUC}^{A|B}_{\text{dir}}$.

Because directional AUC below 0.5 may indicate reversed ranking rather than loss of separability, an orientation-free diagnostic quantity was also reported,

$$\begin{aligned}\text{AUC}_{\text{sep}} &= \max\left(\text{AUC}_{\text{dir}}, 1 - \text{AUC}_{\text{dir}}\right)\\ &= 0.5 + \left|\text{AUC}_{\text{dir}} - 0.5\right|.\end{aligned} \tag{21}$$

AUC_{sep} characterizes rank separability only and is not deployable performance, because post-hoc score inversion would require target-label information. The fixed-orientation transfer gap was

$$\Delta_{\text{transfer}} = \text{AUC}^{\text{within}}_{A|B} - \text{AUC}^{A|B}_{\text{dir}}, \tag{22}$$

and therefore reflects degradation of learned score orientation rather than loss of orientation-free separability.

Recording-related nuisance variation was assessed separately. Acquisition timestamps were recovered for 287 of 288 runs across seven recording dates. The only duplicated fault–condition state recorded twice was used as a nuisance control, and same-date compound pairs sharing one constituent were compared under matched operating conditions. These analyses reduced recording-date mismatch but were treated as sensitivity controls rather than causal estimates of acquisition-session effects.

A complementary feature-level analysis examined how predefined physical signatures changed when constituent A was embedded in a compound fault. Only mechanism-specific curated features defined independently of the cross-partner results were used.

For each curated feature j of constituent A, the reference direction was estimated exclusively from matched healthy and single-fault runs:

$$d_j^A = \text{sign}^+ \left[\text{median}_c \left(x^A_{j,c} - x^H_{j,c}\right)\right], \tag{23}$$

where c indexes matched operating conditions, $x^H_{j,c}$ and $x^A_{j,c}$ denote the healthy and single-fault feature values, respectively, and $\text{sign}^+(0) = +1$. A robust healthy-reference scale s_j^A was estimated from $\{x^H_{j,c}\}_c$ using 1.4826 times the median absolute deviation. If this estimate was degenerate, the interquartile range divided by 1.349 and then the standard deviation were used sequentially; if all remained degenerate, the deterministic fallback $\max(10^{-3}\,\text{median}_c\,|x^H_{j,c}|, 10^{-9})$ was used. No compound observation was used to determine d_j^A or s_j^A. When multiple recordings shared the same original fault label and operating condition, their numeric physics features were median-aggregated before forming the matched reference.

The directed standardized evidence at operating condition c was

$$\delta^A_{j,c} = d_j^A \frac{x^A_{j,c} - x^H_{j,c}}{s_j^A}, \qquad \delta^{A|B}_{j,c} = d_j^A \frac{x^{A|B}_{j,c} - x^H_{j,c}}{s_j^A}, \tag{24}$$

where $x^{A|B}_{j,c}$ denotes the corresponding feature value when constituent A occurs with partner B. Feature preservation was then defined as

$$\rho^{A|B}_{j,c} = \frac{\delta^{A|B}_{j,c}}{\delta^A_{j,c}}. \tag{25}$$

Preservation ratios were evaluated only when $|\delta^A_{j,c}| \geq 0.10$ to avoid unstable ratios for negligible single-fault evidence. For the reported sign-reversal and severe-suppression rates, the analysis was further restricted to feature–condition observations with positive single-fault evidence ($\delta^A_{j,c} > 0$). Negative preservation values indicate feature-level sign reversal, values between zero and one suppression, values near one preservation, and values above one amplification. Severe suppression was defined as $\rho^{A|B}_{j,c} < 0.5$. The analysis characterizes feature-level heterogeneity and is not interpreted as a causal decomposition of the multivariate classifier score.

H. Statistical Analysis

Uncertainty of the principal constituent-level metrics was estimated by nonparametric cluster bootstrap at the physical-run level. In the one-sided regime, both directional predictions from the same run were retained within the same bootstrap cluster. Micro-F_1 was treated as the principal aggregate metric, and the recombination-versus-two-sided difference-of-differences as the principal context-by-condition contrast. Constituent recall and R_{all} were supportive secondary outcomes. Because R_{all} does not penalize false additions, it was interpreted jointly with exact match and FA/run rather than as a stand-alone measure. R_{any} was retained only descriptively because an input-independent predictor can attain one. Other mechanism-specific, isolated-fault-learnability, threshold, classifier-family, and transfer analyses were interpreted as descriptive, sensitivity, or mechanistic analyses and were not used for post-hoc selection of the primary model or operating point.

Because the same physical compound runs contributed to the compared context regimes, paired bootstrap contrasts were evaluated separately under seen and unseen operating conditions:

$$\Delta_{\text{R-O}} = M_{\text{recomb}} - M_{\text{one}}, \tag{26}$$

$$\Delta_{\text{R-T}} = M_{\text{recomb}} - M_{\text{two}}, \tag{27}$$

$$\Delta_{\text{O-T}} = M_{\text{one}} - M_{\text{two}}, \tag{28}$$

where M denotes the metric of interest. Context-by-condition interaction was quantified as

$$\Delta_{\text{int}} = \Delta_{\text{unseen}} - \Delta_{\text{seen}}. \tag{29}$$

Run-level confidence intervals used 5000 paired cluster-bootstrap resamples and the 2.5th and 97.5th percentiles; an effect was considered resolved when the interval excluded zero.

Because the 108 compound runs represent only nine distinct compositions, the recombination-versus-two-sided interaction was additionally subjected to a composition-level cluster-bootstrap sensitivity analysis. Each of 10,000 replicates sampled the nine compositions with replacement while retaining all 12 operating-condition runs of each sampled composition and applying identical composition multiplicities to the seen and unseen cells. Sensitivity to individual compositions was also examined by leave-one-composition-out analysis.

Closed-set confidence intervals used 5000 stratified run-level bootstrap resamples within the 24 diagnostic classes.

For cross-partner transfer, score-orientation reversal was tested using

$$T = \left|\text{AUC}_{\text{dir}} - 0.5\right|. \tag{30}$$

Directional-AUC confidence intervals used 2000 bootstrap resamples clustered by operating condition. For the orientation test, positive and negative scores were swapped within

TABLE III: Closed-set multiclass diagnostic performance with 95% bootstrap confidence intervals.

Metric	Estimate	95% CI
Accuracy	0.9132	[0.8854, 0.9375]
Balanced accuracy	0.9134	[0.8856, 0.9377]
Macro-F_1	0.9119	[0.8826, 0.9366]

matched operating conditions over 5000 permutation replicates using $|\mathrm{AUC}_{\mathrm{dir}} - 0.5|$ as the test statistic. Family-wise multiplicity across the 18 directed constituent–partner relations was controlled by the Holm procedure separately within each condition-mode family. A relation was classified as a statistically supported score-orientation reversal only when its directional AUC and bootstrap interval were below 0.5 and the Holm-adjusted permutation test supported departure from chance orientation.

IV. Results

A. Closed-Set Discrimination and the Generalization Gap

The auxiliary 24-class closed-set classifier achieved an accuracy of 0.9132, balanced accuracy of 0.9134, and macro-F_1 of 0.9119 (Table III), indicating strong conventional in-distribution separability.

The substantially lower constituent-level performance under the crossed protocols is therefore consistent with the additional demands of composition, compound-context, and operating-condition generalization rather than poor baseline separability alone. Because the closed-set reference uses a different model architecture from the locked constituent probe, this comparison is descriptive rather than a model-controlled estimate of task difficulty; classifier-family robustness analyses below separately assess the contribution of constituent-model choice.

B. Crossed Composition–Context–Condition Generalization

The crossed protocol was evaluated on all 108 physical compound-fault runs under recombination, one-sided, and two-sided context regimes, each with seen and unseen operating conditions. Both directional removals were evaluated in the one-sided regime. Table IV summarizes the principal constituent-level results.

Recombination yielded the strongest overall performance, with micro-F_1 of 0.6775 and 0.6865 and constituent recall of 0.6759 and 0.6944 under seen and unseen conditions, respectively. Exact constituent-set recovery nevertheless remained low across all six regimes (0.0833–0.1574), indicating that partial recovery was substantially easier than exact decomposition.

The prevalence references in Table V show why partial-recovery metrics cannot be interpreted in isolation. A constant inner-plus-outer predictor achieves $R_{\mathrm{any}} = 1$, while adding broken bar raises constituent recall to 0.6667 but produces 1.6667 false additions per run and no exact matches. By comparison, the locked detector achieved micro-F_1 of 0.631–0.686 with FA/run of 0.639–0.732. Under recombination, R_{all} reached 0.4167 and 0.4444, compared with 0.3333 for the constant inner-plus-outer-plus-broken-bar reference. That reference, however, incurred 1.6667 false additions per run, compared with 0.6389 and 0.6574 for the learned detector. Thus, the advantage of the learned detector is a more favorable joint recovery–overdiagnosis trade-off rather than uniform superiority on every individual metric.

C. Mechanism-Specific Failure Modes: Isolated Learnability, Compound Degradation, and Model Capacity

Table VI reveals that poor compound decomposition does not arise from a single common failure mechanism. Bearing-inner, bearing-outer, and broken-bar constituents were highly discriminable both in isolation and in compound evaluation. Their compound AUROCs remained between 0.965 and 0.993, and their locked F_1 values substantially exceeded the corresponding input-independent references. Thus, their strong thresholded performance reflects genuine constituent discrimination rather than prevalence alone.

Dynamic eccentricity showed the clearest compound-context degradation. It was almost perfectly discriminable from other non-compound states under leave-one-condition-out evaluation (AUROC=0.9995; $F_1 = 0.923$), yet its compound recall fell to 0.017 and its locked compound F_1 to 0.028. Random forest increased compound F_1 to 0.385, indicating additional classifier-capacity sensitivity, but recovery remained modest. The combination of strong isolated learnability and severe compound degradation is consistent with a pronounced loss of decision consistency in the compound setting and shows that the poor compound result cannot be explained by weak isolated-fault detectability alone.

Static eccentricity exhibited a different failure mode. It was also strongly learnable in isolation (AUROC=0.992; $F_1 = 0.920$) and retained substantial ranking information in compound evaluation (AUROC=0.867), despite only moderate locked F_1 (0.577). The RBF-SVM increased descriptive compound F_1 to 0.866 under the same crossed protocol, indicating pronounced classifier-family sensitivity rather than absence of discriminative information in the curated representation.

Winding differed from both eccentricity mechanisms. Its isolated-fault specificity was already weaker (AUROC=0.823; $F_1 = 0.400$), and compound performance deteriorated further ($F_1 = 0.205$). None of the nonlinear alternatives improved on the locked linear probe, while its mean compound AUROC was below chance (0.398). This aggregate constituent-level AUROC is distinct from the directed cross-partner transfer analysis below and is therefore not interpreted as evidence of a statistically supported score-orientation reversal. Rather, it indicates poor fixed-score ranking on average across the crossed scenarios. Together, these results reveal three distinct empirical failure patterns: severe compound degradation for dynamic eccentricity, strong classifier-family sensitivity for static eccentricity, and weaker constituent specificity with persistent compound difficulty for winding.

D. Paired Effects of Compound-Context Deprivation

The principal comparison was the recombination-versus-two-sided context-by-condition interaction in micro-F_1, with constituent recall and R_{all} as supportive recovery outcomes.

TABLE IV: Constituent-level performance across context and operating-condition regimes. EM denotes exact constituent-set recovery, R_{const} constituent recall, FA/run false additions per prediction row, and R_{any} and R_{all} recovery of at least one and all true constituents, respectively. Boldface denotes the descriptively best value within each operating-condition block and does not imply a statistically resolved difference.

Condition	Context regime	Active Macro-F_1	Micro-F_1	EM	R_{const}	FA/run	R_{any}	R_{all}
Seen	One-sided	0.5890	0.6581	0.1204	0.6505	0.6528	0.9259	0.3750
Seen	Recombination	**0.6177**	**0.6775**	0.1389	**0.6759**	**0.6389**	0.9352	**0.4167**
Seen	Two-sided	0.5994	0.6620	0.1111	0.6620	0.6759	**0.9444**	0.3796
Unseen	One-sided	0.5761	0.6452	0.1111	0.6481	0.7222	0.9398	0.3565
Unseen	Recombination	**0.6177**	**0.6865**	**0.1574**	**0.6944**	**0.6574**	**0.9444**	**0.4444**
Unseen	Two-sided	0.5588	0.6311	0.0833	0.6296	0.7315	0.9352	0.3241

TABLE V: Input-independent constant-predictor references on the 108 compound runs, quantifying performance obtainable from constituent prevalence alone.

Constant prediction	Micro-F_1	EM	R_{const}	FA/run	R_{any}	R_{all}
Inner + outer	0.5556	0.1111	0.5556	0.8889	1.0000	0.1111
Inner + outer + broken bar	0.5333	0.0000	0.6667	1.6667	1.0000	0.3333
Inner only	0.3704	0.0000	0.2778	0.4444	0.5556	0.0000

TABLE VI: Mechanism-specific diagnostic decomposition. Single-fault AUROC and F_1 are from leave-one-operating-condition-out discrimination of the target single fault against all other non-compound states. Compound AUROC, recall, and LR F_1 are averaged equally across the six crossed scenarios. Const.-F_1 is the constituent-specific input-independent always-positive reference. RF, RBF-SVM, and MLP results are descriptive classifier-family sensitivities and were not used for model selection.

Constituent	Single AUROC	Single F_1	Compound AUROC	Compound Recall	LR F_1	Const. F_1	RF F_1	RBF-SVM F_1	MLP F_1
Bearing inner	0.999	0.957	0.965	0.896	0.940	0.714	0.947	0.839	0.936
Bearing outer	1.000	1.000	0.988	0.796	0.886	0.714	0.898	0.900	0.874
Broken bar	1.000	1.000	0.993	0.927	0.923	0.364	0.919	0.883	0.906
Dynamic eccentricity	1.000	0.923	0.590	0.017	0.028	0.364	0.385	0.054	0.085
Static eccentricity	0.992	0.920	0.867	0.538	0.577	0.364	0.658	0.866	0.814
Winding	0.823	0.400	0.398	0.229	0.205	0.364	0.019	0.163	0.115

Paired run-cluster bootstrap contrasts are summarized in Fig. 2.

Under seen operating conditions, recombination showed only limited advantages. For constituent recall, the recombination–one-sided difference was 0.0255 (95% CI: $[0.0000, 0.0509]$), whereas the recombination–two-sided difference was 0.0139 (95% CI: $[-0.0186, 0.0509]$); the corresponding micro-F_1 intervals also included zero.

Under unseen conditions, the recombination advantage increased. Relative to one-sided context zero-shot, recall improved by 0.0463 (95% CI: $[0.0231, 0.0694]$) and micro-F_1 by 0.0413 (95% CI: $[0.0223, 0.0636]$). Relative to two-sided zero-shot, the corresponding improvements were 0.0648 ($[0.0278, 0.1065]$) and 0.0554 ($[0.0232, 0.0895]$), respectively. One-sided–two-sided differences remained unresolved for both metrics.

At the physical-run level, the recombination-versus-two-sided difference-of-differences was 0.0400 for micro-F_1 (95% CI: $[0.0088, 0.0731]$), 0.0509 for constituent recall ($[0.0185, 0.0880]$), and 0.0833 for R_{all} ($[0.0278, 0.1481]$).

Because the 108 compound runs represent only nine distinct compositions, the interaction was also evaluated by composition-level cluster bootstrap. Under this stricter analysis, the micro-F_1 interaction remained positive but unresolved ($\Delta_{\mathrm{int}} = 0.0400$, 95% CI: $[-0.0035, 0.0861]$), whereas constituent recall (0.0509, $[0.0093, 0.0926]$) and R_{all} (0.0833, $[0.0093, 0.1574]$) remained resolved. The micro-F_1 interaction was positive in all nine leave-one-composition-out analyses (0.0236–0.0527), indicating stable direction but composition-level uncertainty in its magnitude.

Because recombination retained more training runs than two-sided withholding, the recombination training sets were repeatedly downsampled to match the corresponding two-sided training-set cardinalities. Matching was exact in both condition regimes. Across 30 repeated subsamples, the unseen-condition recombination advantage remained positive on average for micro-F_1 (+0.0335), constituent recall (+0.0438), and R_{all} (+0.0778). However, composition-level bootstrap intervals included zero for micro-F_1 ($[-0.0151, 0.0813]$), constituent recall ($[-0.0046, 0.0880]$), and R_{all} ($[-0.0185, 0.1667]$).

The corresponding context-by-condition interactions also remained positive on average but unresolved: $+0.0105$ for micro-F_1 (95% CI $[-0.0401, 0.0606]$), $+0.0205$ for constituent recall ($[-0.0370, 0.0880]$), and $+0.0312$ for R_{all} ($[-0.0741, 0.1574]$). Thus, the directional pattern of the primary analysis was retained after matching total training-set cardinality, but the interaction was not statistically resolved under this sensitivity analysis. Accordingly, the primary context effect should not be interpreted as independent of available training support.

The recombination-versus-one-sided interaction was unresolved at the run-cluster level for micro-F_1 and recall, although R_{all} showed a positive interaction of 0.0463 (95% CI: $[0.0046, 0.1019]$). In the primary analysis, complete compound-context deprivation had a larger adverse effect under operating-condition shift. However, after matching total training-set cardinality, the recombination-versus-two-sided interaction remained directionally positive but unresolved for all three principal recovery outcomes.

One-sided directional results were also heterogeneous across the directional withholding subsets. When broken-bar compound context was withheld, aggregate constituent recall within the corresponding test subset was 0.979 and 0.896 under seen and unseen conditions, respectively. The corresponding subset-level values were 0.438 and 0.479 when winding context was withheld and 0.500 under both condition settings when dynamic-eccentricity context was withheld. These values are aggregate recall over all true constituents in the affected compound runs and should not be interpreted as recall of the withheld constituent itself. In particular, the mechanism-specific analysis in Table VI shows that mean dynamic-eccentricity recall was only 0.017 across the crossed scenarios, despite near-perfect isolated-fault discrimination. Both directional predictions from each physical run remained within the same bootstrap cluster.

E. Sensitivity to Decision-Threshold Objective

Post-hoc threshold calibration with $F_{0.5}$ and balanced accuracy was used to assess operating-point sensitivity. Relative to the locked F_1 objective, $F_{0.5}$ reduced FA/run from 0.639–0.732 to 0.222–0.389 and increased exact-match recovery from 0.083–0.157 to 0.157–0.287, while reducing constituent recall to 0.583–0.644. Balanced-accuracy calibration shifted the trade-off toward higher recall and more false additions. Thus, exact-set recovery was substantially more operating-point sensitive than aggregate recall, and the locked F_1 operating point should not be viewed as an upper bound on achievable decomposition performance.

F. Sensitivity to Single-Fault Severity Merging

Removing low-severity single-fault support did not produce a uniform change in constituent recall, exact-match recovery, or all-constituent recovery across the crossed regimes. Most paired differences for these metrics had 95% confidence intervals spanning zero, indicating that the principal generalization findings were not driven by pooling low- and high-severity single-fault examples.

The clearest differences were observed in false additions. Under seen conditions, removing low-severity support increased FA/run from 0.653 to 0.755 in the one-sided regime ($\Delta = +0.102$, 95% CI: $[0.046, 0.162]$) and from 0.639 to 0.722 under recombination ($\Delta = +0.083$, 95% CI: $[0.019, 0.157]$). Under unseen conditions, the two-sided regime similarly increased from 0.731 to 0.843 FA/run ($\Delta = +0.111$, 95% CI: $[0.037, 0.185]$).

For seen-condition recombination, micro-F_1 decreased from 0.677 to 0.651 when low-severity support was removed ($\Delta = -0.026$, 95% CI: $[-0.052, -0.002]$). Other micro-F_1 differences were unresolved. Overall, the sensitivity analysis indicates that severity merging does not account for the principal crossed-generalization pattern and, in several regimes, the additional low-severity support reduced over-diagnosis rather than artificially inflating constituent recovery.

TABLE VII: Descriptive classifier-family robustness averaged across the six crossed evaluation scenarios. LR denotes logistic regression and MLP denotes the three-hidden-layer feed-forward neural baseline.

Model	Micro-F_1	EM	R_{const}	FA/run
LR	0.660	0.120	0.660	0.680
RF	0.744	0.282	0.669	0.256
RBF-SVM	0.692	0.157	0.664	0.509
MLP	0.737	0.310	0.659	0.256

G. Robustness Across Classifier Families

The crossed protocol was repeated with random forest, RBF-SVM, and MLP classifiers as post-hoc robustness analyses. Table VII summarizes descriptive performance averaged across the six crossed scenarios; logistic regression remained the locked primary model.

Absolute decomposition performance varied substantially by classifier family: RF achieved the highest mean micro-F_1 (0.744), MLP the highest exact-match rate (0.310), and both reduced mean FA/run to 0.256, while constituent recall remained similar across models (0.659–0.669). Nevertheless, the unseen-condition recombination advantage over two-sided context zero-shot remained positive for all four classifiers (micro-F_1: 0.023–0.055; constituent recall: 0.032–0.069), indicating that the qualitative context effect was not specific to the locked linear probe.

H. Oracle Detectability and Cross-Partner Constituent Transfer

Within-pair analysis first tested whether constituent evidence remained measurable in the target compound. Across all 18 directed relations $A|B$, condition-held-out AUC for the contrast $A + B$ versus B ranged from 0.9167 to 1.000, with a median of 1.000. Thus, target compounds remained strongly separable from their partner-only references.

However, recording-related nuisance variation was also detectable. The two available recordings of the same bearing-outer fault under the same operating condition were nearly perfectly separable (AUC $=$ 0.998), indicating that high within-pair AUC cannot be attributed uniquely to the added constituent.

A same-date sensitivity control reduced recording-date mismatch. Eight same-date compound pairs sharing one constituent and all 12 operating conditions yielded 16 directed comparisons, with leave-one-condition-out AUCs of 0.972–1.000. Static eccentricity in the bearing-outer context remained perfectly separable from same-date alternatives containing either broken bar or winding (AUC $=$ 1.000). These controls do not causally isolate the target mechanism, but they show

TABLE VIII: Selected directed cross-partner transfer results. Within-pair AUC quantifies detectability of constituent A in the target pair, whereas cross-partner AUC quantifies transfer of evidence learned from other compound partners.

Target composition	Constituent A	Partner B	Within-pair AUC	Cross-partner AUC (seen)	Cross-partner AUC (unseen)
Bearing inner + bearing outer	Bearing outer	Bearing inner	1.000	1.000	1.000
Bearing inner + dynamic eccentricity	Bearing inner	Dynamic eccentricity	1.000	0.944	0.931
Bearing inner + broken bar	Broken bar	Bearing inner	1.000	0.893	0.851
Bearing inner + dynamic eccentricity	Dynamic eccentricity	Bearing inner	1.000	0.711	0.653
Bearing outer + static eccentricity	Static eccentricity	Bearing outer	0.917	0.000	0.000
Bearing outer + winding	Winding	Bearing outer	0.917	0.333	0.340

that compound-state separability cannot be explained solely by recording-date mismatch.

Cross-partner transfer therefore tested a stricter question: whether evidence for constituent A learned from other compound contexts, such as $A + C$ versus C, generalized to the target contrast $A + B$ versus B. Under seen conditions, directional cross-partner AUC across the 18 relations had a median of 0.905 and a mean of 0.815; 13 relations achieved AUC $\geq$ 0.80 and 15 achieved AUC $\geq$ 0.70. Under unseen conditions, the median and mean remained 0.884 and 0.803, respectively. Selected results are shown in Table VIII, and the full directional pattern in Fig. 3.

Transfer was nevertheless strongly heterogeneous. For static eccentricity with bearing outer, within-pair AUC was 0.9167, whereas directional cross-partner AUC was 0.000 under both seen and unseen conditions. Because $\mathrm{AUC}_{\mathrm{sep}} = 1.000$, the target remained perfectly rank separable but in the opposite orientation. This was the only relation with a statistically supported score-orientation reversal after multiplicity correction (Holm-adjusted $p = 0.0136$ seen; $p = 0.0108$ unseen). Given the recording separability observed above, this result is interpreted as reproducible context-dependent score reorientation rather than causal proof of physical signature inversion.

Winding with bearing outer also produced below-chance directional AUCs (0.3333 seen; 0.3403 unseen), but orientation-free separability was only approximately 0.67 and the Holm-adjusted tests were not significant. Accordingly, this relation was classified as orientation-ambiguous rather than as a supported reversal.

Overall, compound states remained strongly discriminable while score orientation learned from other compound partners could fail to transfer. Because recording-specific structure was also detectable, the result is interpreted as context-dependent score transfer rather than causal evidence of physical constituent-signature inversion. Figure 4 contrasts within-pair detectability with directional cross-partner transfer.

Fig. 4: Within-pair detectability versus directional cross-partner transfer. The dashed line denotes chance-level ranking. Below-chance values require the separate score-orientation analysis in Section III-H before being classified as supported reversals.

I. Physical-Signature Heterogeneity and Directional Failure Modes

Feature-level preservation showed that score reorientation was not equivalent to uniform inversion of the underlying physical descriptors. For static eccentricity with bearing outer, the median preservation ratio was 0.760 over the full ratio-eligible set; among ratio-eligible observations with positive single-fault evidence, 17.8% showed sign reversal and 34.6% severe suppression. Conversely, bearing-outer evidence in the bearing-inner compound had a lower median preservation ratio ($\rho = 0.305$); within the corresponding positive-single-evidence subset, 18.4% of observations showed sign reversal and 58.4% severe suppression, despite a cross-partner AUC of 1.000. Thus, feature suppression and multivariate score reorientation represent distinct partner-dependent phenomena (Fig. 5).

The locked probe also produced 0.64–0.73 false additions per prediction row, indicating a material over-diagnosis burden despite partial constituent recovery. Recording-related nuisance variation further limits physical attribution: distinct recordings of the same fault state were strongly separable, while recording date provides only a coarse proxy for experimental session. Accordingly, the cross-partner results support context-dependent score transfer but not causal attribution to physical signature inversion.

Overall, the experiments reveal complementary forms of generalization failure. Mechanism-specific analysis shows that otherwise learnable constituents can degrade sharply in compound evaluation, with substantially different patterns across mechanisms. The crossed protocol additionally suggests that complete compound-context deprivation is more detrimental when the operating condition is also unseen in the primary analysis, although this interaction becomes unresolved after matching total training-set cardinality. Finally, cross-partner analysis shows that constituent scores can remain strongly separable yet fail to preserve their learned orientation across partners. These observations describe related but distinct limitations of constituent generalization rather than a single uniform failure mode.

V. Conclusion

This paper evaluated whether the elementary mechanisms of a compound motor fault can be recovered when their exact combination, their compound context, and the operating condition are varied independently of the training history. On a public multimodal induction-motor dataset whose compound faults span mechanical and electrical modalities, a locked linear constituent probe over mechanism-specific physical features showed that strong closed-set discriminability (24-class accuracy 0.913, 95% CI $[0.885, 0.938]$) coexists with exact constituent-set recovery of only 0.083–0.157 on unseen compounds. The two results are not in tension; they measure different abilities, and the gap between them is the paper's subject.

Four conclusions follow from the controlled analyses. First, at the aggregate level, the dominant failure mode is incomplete decomposition rather than complete diagnostic silence. Constituent recall remained between 0.630 and 0.694, but a prevalence-only constant predictor could itself reach 0.667 recall at the cost of 1.667 false additions per run and zero exact matches. Under recombination, all-constituent recovery reached 0.417–0.444, compared with 0.333 for the reported three-constituent prevalence reference; however, that reference incurred 1.667 false additions per run compared with approximately 0.64–0.66 for the learned detector. These results show why constituent recall must be interpreted jointly with false additions, micro-F_1, and exact-set recovery rather than as a stand-alone measure of diagnostic skill. The constituent-resolved controls further showed that these failures do not share a common origin. Dynamic eccentricity was almost perfectly learnable as an isolated fault (AUROC 0.9995; $F_1 = 0.923$) but collapsed to compound recall of 0.017 and F_1 of 0.028, providing the clearest evidence that poor compound recovery cannot be explained by weak isolated detectability alone. Static eccentricity was likewise highly learnable in isolation and retained substantial compound ranking information, while its F_1 increased from 0.577 with the locked linear probe to 0.866 with RBF-SVM, consistent with pronounced classifier-family sensitivity. Winding showed a third pattern: isolated-fault specificity was already weaker and none of the nonlinear alternatives rescued its poor compound performance. In contrast, bearing-inner, bearing-outer, and broken-bar evidence remained strongly discriminative across both isolated and compound settings. Thus, constituent-level failure in this dataset exhibits distinct empirical patterns consistent with compound degradation, classifier-capacity sensitivity, and limited constituent specificity rather than a uniform cross-modal deficit. Second, the adverse effect of compound-context deprivation was larger when the operating condition was simultaneously unseen in the primary analysis. For the recombination-versus-two-sided interaction, composition-level clustering yielded $+0.040$ for micro-F_1 (95% CI $[-0.004, 0.086]$), $+0.051$ for constituent recall (95% CI $[0.009, 0.093]$), and $+0.083$ for all-constituent recovery (95% CI $[0.009, 0.157]$). Thus, the primary composition-level analysis supported the interaction for constituent recall and all-constituent recovery, while the micro-F_1 interaction remained unresolved. After matching recombination training-set cardinality to the two-sided regime, the interactions remained positive on average but were unresolved for micro-F_1 ($+0.010$, 95% CI $[-0.040, 0.061]$), constituent recall ($+0.021$, $[-0.037, 0.088]$), and all-constituent recovery ($+0.031$, $[-0.074, 0.157]$). The observed context effect is therefore sensitive to available training support and should not be interpreted as a training-cardinality-independent effect of compound-context exposure. Third, constituent evidence transfers across compound partners often but directionally: within-pair detectability was near-perfect throughout, yet cross-partner transfer ranged from perfect to a statistically supported score-orientation reversal (Holm-adjusted $p \leq 0.014$)—and the demonstrated separability of distinct recordings of the same fault state requires that reversal to be interpreted as context-dependent score transfer rather than as proven physical signature inversion. Fourth, absolute decomposition quality is an operating-point and model property: precision-weighted thresholding raised exact-match recovery to 0.157–0.287 at reduced recall, and nonlinear classifier families reduced false additions from $\sim$0.68 to $\sim$0.26 per run while the unseen-condition recombination advantage over two-sided withholding remained directionally positive across all four classifier families.

For practice, these results caution against reading closed-set accuracy as deployment readiness, argue for calibrating constituent detectors with explicit precision–recall cost structure, and indicate that commissioning data should sample compound contexts and operating conditions jointly rather than relying on either alone. Several directions remain open. The dataset provides insufficient replicated same-state recordings to separate fault, assembly, sensor-mounting, thermal, and recording-related effects. The transfer analyses therefore stop at carefully bounded attribution; systematically replicated acquisitions would enable stronger separation of these potential sources of variation. Representation or normalization strategies that promote constituent-score consistency across compound partners are identified as hypotheses, not established remedies. Finally, the constituent-level protocol, prevalence references, and controls used here are directly portable to the companion gearbox release and to other multimodal datasets, where cross-dataset replication of the context-by-condition interaction would test its generality.

DATA AVAILABILITY

The dataset analyzed in this study is publicly available from Mendeley Data at https://doi.org/10.17632/6s3dggj9mw.1 under a CC BY 4.0 license [34], with its accompanying descriptor article [1]. The acquisition-timestamp metadata used for the recording-sensitivity and same-date controls were recovered directly from the released file names and therefore require no additional data source to reconstruct.

CODE AVAILABILITY

The code developed and used during the current study is available from the corresponding author upon reasonable request.

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Fig. 1: Graph interpretation of compound-context withholding. Nodes denote constituent mechanisms and edges observed compound compositions. Recombination removes only the target edge, one-sided zero-shot removes compound exposure for one target constituent, and two-sided zero-shot removes it for both.

Fig. 2: Paired run-cluster bootstrap differences between context regimes. Points denote differences in constituent recall and micro-F_1; horizontal bars denote 95% bootstrap confidence intervals. Positive values favor the first regime in each contrast.

Fig. 3: Directional cross-partner transfer AUC. Rows denote constituent A and columns partner B. Values below 0.5 indicate below-chance fixed-orientation ranking but do not alone establish a statistically supported score-orientation reversal. n_C denotes the number of eligible compound training contexts for the recovered constituent.

Fig. 5: Feature-level preservation of selected static-eccentricity signatures under bearing-inner and bearing-outer partner contexts. Features were selected by prioritizing sign reversals and then partner-dependent preservation differences; aggregate preservation ratios were calculated over the full ratio-eligible feature–condition set, whereas sign-reversal and severe-suppression rates were calculated only over ratio-eligible observations with positive single-fault evidence. Negative values indicate sign reversal, values near one preservation, and values above one amplification.